



Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}$$



Monge's Problem:

(Discrete measure version)

$$\min_{T_{\#}\alpha=\beta} \frac{1}{n} \sum_{i=1}^n c(x_i, T(x_i))$$

Re-visting Discrete Monge

Data indexed by

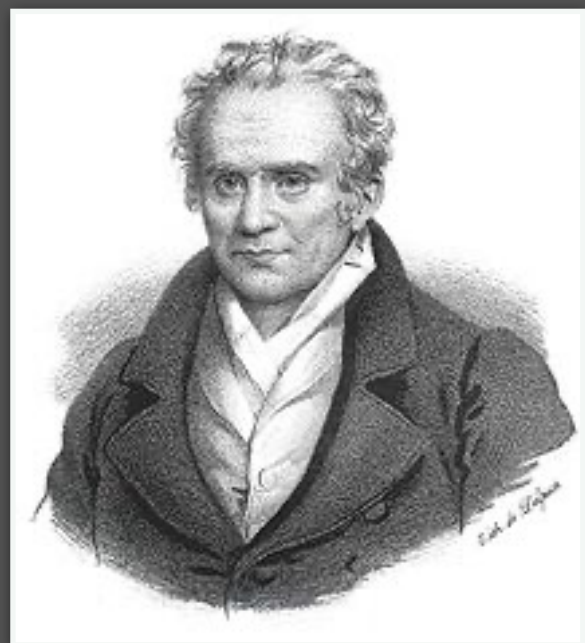
$$i \in \{1, \dots, n\}, j \in \{1, \dots, n\}$$

Cost matrix: $(C_{i,j})_{i,j=1}^n \in \mathbb{R}^{n \times n}$

Search over permutations in

$$\Sigma_n = \{\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}\}$$

We want the min-cost one



Monge's Problem:
(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}$$

Data expressed as empirical measures

$$\alpha = \sum_{i=1}^n \mathbf{a}_i \delta_{x_i}, \beta = \sum_{j=1}^n \mathbf{b}_j \delta_{y_j}$$

Costs: $c(x_i, y_j)$ (can also be written as matrix)

Search over pushforward maps from

α to β

We want the min-cost one



Monge's Problem:
(Discrete measure version)

$$\min_{T_{\#}\alpha=\beta} \frac{1}{n} \sum_{i=1}^n c(x_i, T(x_i))$$

Monge Formulation - General